

Lab 2

Q1 Insert a new faculty called 'Faculty of Law' with dean = 'Dr. Hany Aziz', building = 'G', budget = 8,000,000. Return the new faculty_id immediately.

Q2 Give all professors in department 3 a 15% salary raise. Return each professor's name, old salary, and new salary.

Q3 Update all students whose GPA is below 2.0 and who enrolled before 2022 — set is_active to FALSE. Return their names.

Q4 Enroll student_id 5 (Khaled) in course_id 1 (CS101) for Fall 2023. If the enrollment already exists, do nothing.

Q5 Update the grade of student_id 1 in course_id 3 for Fall 2022 to 98, and set letter_grade to 'A+'.

Q6 Use MERGE (PG 15+): if student_id=99 exists, update their address; if not, insert them as a new student.

Q7 Create a new table called high_gpa_students containing only students with GPA >= 3.5, using SELECT INTO.

Q8 Create a table dept_summary containing: dept_name, student count, average GPA, and total scholarship amount per department — using CREATE TABLE AS.

Q9 Make two copies of the enrollments table: one with structure only (no data), and another with structure + data + all constraints.

Q10 Create a table exam_results with various DEFAULT values: status='pending', score=0, exam_date=CURRENT_DATE, created_by=CURRENT_USER. Insert two rows — one using all defaults, one overriding them.

Q11 Show all students who have metadata stored. Display: their first hobby, the number of languages they speak, and whether they have a laptop.

Q12 Create an ENUM type called student_level with values: Freshman, Sophomore, Junior, Senior. Add it as a column on students and update it based on GPA ranges.

Q13 Create a Composite Type called contact_info with fields (phone TEXT, email TEXT, city TEXT). Use it in a student_contacts table.